

PENGYU JIE

Rouen, France Email: mamba579jpy@gmail.com Phone: +33 7 73 12 40 83

Portfolio: <https://rajayoux.github.io> GitHub: <https://github.com/JayGarland>

Available from Feb 2026 for a paid full-time internship (4–6 months) with a signed internship agreement (convention de stage). On-site or hybrid; not 100% remote (max 2 telework days/week).

French residence permit valid

Summary

Unity + XR developer focused on turning prototypes into stable, user-testable builds with a builder's mindset and shipping focus. Comfortable across MRTK/Meta XR with clean scene/prefab organization, input/camera rigs, and content packaging (Addressables/streaming).

Skills

Engines/SDKs: Unity (2020/2022), OpenXR, XR Interaction Toolkit (XRI), MRTK, Meta XR

Languages: C#, Python, C, SQL, JavaScript, Java

Tools: Unity Profiler, Git/GitHub, GitHub Actions

Devices/Platforms: Pico (source), NOLOVR (target), Quest 2/3 (personal dev), Windows/Android (publication)

Experience

VR Applications & Rendering Intern — Hunantv.com Interactive Entertainment Media Co., Ltd.

MangoTV VR — Device Porting & XR Integration (Dec 2023 – Apr 2024)

Role: Unity XR Developer (Large-scale team: VR dept. 15+, company 5000+)

Tech: Unity (C#), XRI, Addressables/Streaming, HybridCLR, Unity Profiler, NOLOVR SDK, Meta XR SDK, Scene Understanding

- Ported from Pico→NOLOVR with input remapping/camera rig adaptation; implemented Addressables/streaming architecture.
- Practiced manager-class architecture, team protocols, profiling for multi-device deployment; explored HybridCLR hot-update.
- R&D on Meta XR scene understanding/MR for “virtual show” concepts; prototyped physics-based virtual objects with occlusion/zoom.

Outcome: Faster loading and stable interaction mapping; iterative test builds matched production constraints.

Software Engineering Assistant — Huapu Zhilian (Beijing) Technology Co., Ltd.

Jul 2022 – Aug 2022 • Beijing, China

Project prototypes in Unity; internship listed on my site: rajayoux.github.io

Selected Projects

EchoChamber (回音殿) — VR Cultural Physics Experience (Apr 2025 – May 2025)

Role: Lead VR Developer & Cultural Technology Designer

Tech: Unity (C#), Meta XR SDK, Spatial Audio, Hand Tracking, Physics Simulation, Room Acoustics

- Built a physics-based 3D audio/room acoustics engine with hand-tracked sound manipulation and spatial placement.
- Integrated traditional Chinese acoustic philosophy into modern VR interaction and education experience.

Outcome: Delivered an innovative VR experience showcasing spatial audio and cross-cultural technical design.

Mixed Reality Huarong Dao (MRPuzzle) — MR Puzzle Game (Jan 2024 – May 2024)

Role: Lead Developer & Designer

Tech: Unity (C#), MRTK, Oculus Integration, Meta Quest 3, Spatial Anchoring, Passthrough API

- Implemented spatial anchoring to bind a virtual board to real desktops; dual input (hand tracking + controllers).
- Used MRTK remote ray/finger poke, Passthrough controls, particle transitions, chaptered gameplay.

Outcome: Delivered a polished MR puzzle with scalable difficulty and dual puzzle modes.

Auto Reply Bot (open-source) — Python • Docker • Cloud (Jul 2023 – Oct 2023)

Parses subreddit posts/comments with prompt templates and context; integrates image recognition and content generation.

170 GitHub stars; CI via GitHub Actions; containerized deployment. Source:

https://github.com/JayGarland/Autoreply_Sydneybot_Reddit

Journey of Chinese Traditional Medicine — VR Education Game (Apr 2023 – Oct 2023)

Role: Lead Developer & Project Manager

Tech: Unity (C#), MRTK v2.73, OpenAI API, Quest 2/Rift, Voice Recognition

- Built herb collection/medicine crafting systems with temperature-based brewing and treatment workflows.
- Integrated AI dialogue with Zhang Zhongjing; optimized VR performance and cross-platform builds.

Outcome: MTE 2023 Gold + Special Award; CCC Contest Provincial First Prize.

Chinese Characters Quest — VR Language Learning (Spring 2022 – Fall 2022)

Role: Lead Developer & Game Designer

Tech: Unity (C#), Oculus Integration, PDollar Cloud Gesture Recognition, Blender

- Gesture-based Chinese character writing with PDollar Cloud; accessible classroom experiences and mini-games.

Outcome: National Second Prize in China Collegiate Computing Contest.

Cyber Renaissance (VR Game) (May – Jul 2022);

UC Berkeley cs61a (Self-study) (Jun – Dec 2022).

Education

ESIGELEC — Master in Sciences & Technology: Software Engineering & Digital Transformation

(Sep 2024 – Jun 2026, current)

HVUST — B.S. Software Engineering

(Sep 2020 – Jun 2024)

Awards

Malaysia Technology Expo (MTE 2023) — Gold + Special Award | 2023-10

China Collegiate Computing Contest — Provincial First Prize (First Author) | 2023-06

China Collegiate Computing Contest — National Second Prize (First Author) | 2022-08

Provincial Excellent Undergraduate Innovation Training Program — Top 10% | 2023-09

Languages

English — Professional working

French — B1 (daily communication)

Chinese — Native